

Balanced-coherence regimes of conscious access: a computational model with public EEG validation

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Short title: Balanced-coherence regimes for neural access

Abstract

The Gamma-Code Consensus (GCC) is a computational model of conscious-access-compatible dynamics. It proposes that global access is not maximized by synchrony alone, but by a bounded regime in which coherence, effective dimensionality, and temporal variability jointly remain within calibrated ranges. We formalize this regime in a stochastic phase-oscillator network with separable topology, global gain, and population-specific efficacy, then derive six falsifiable predictions. Synthetic network simulations reproduce the evaluated prediction family: nonlinear thresholding, degradation-dependent threshold shifts, bounded access intervals, anesthesia-like trajectories, and selective-preservation re-entry under controlled assumptions. Public EEG analyses provide convergent but deliberately limited empirical support. In Chennu et al. 2016 ($N = 20$), the subject-calibrated access index declines under propofol sedation and survives fsaverage template-source reconstruction. In OpenNeuro DS005620 ($N = 20$ paired), the same direction replicates, and frozen cross-dataset gamma benchmarks generalize above chance, including a wPLI-based lagged-coupling variant. Sleep-EDF (22 recordings) shows cross-paradigm state geometry: GCC triad features separate Wake/NREM and REM/NREM, although spectral sleep features remain stronger. A public disorders-of-consciousness EEG/PSG dataset provides a pilot clinical anchor: alpha-band GCC distinguishes MCS+ from VS and remains positive after spectral and retained-epoch

26 residualization. Spectral controls show that GCC is not a bandpower-independent biomarker.

27 The present contribution is therefore a theory-guided, empirically stress-tested regime model

28 of access-compatible dynamics; the stronger clinical hypothesis of selective re-entry in struc-

29 turally degraded brains remains a future target.

30 **Keywords:** consciousness; global access; EEG; propofol; disorders of consciousness; metasta-
31 bility; phase synchronization; computational neuroscience

Introduction

1. Aim and Scope

The GCC is a principled model of level of consciousness and global access. It asks when a heterogeneous, noisy, and degradable network enters an integrated but non-rigid macroscopic regime. It is not a theory of qualia; it is a regime model related to global-access, integration/differentiation, metastability, perturbational-complexity, and clinical state-marker approaches [3–8, 11, 13, 15].

The evaluated core claim is that access-compatible dynamics occupy a bounded region jointly characterized by coherence, effective dimensionality, and temporal variability. This claim generates predictions P2–P6 and is tested through synthetic simulations and public EEG/PSG analyses. A narrower degradation extension concerns selective re-integration in damaged systems:

In degraded neural systems, global access can arise not only from increased activity but also from selective collapse of competing parts of the network. Functional re-integration then follows from gain redistribution onto a residual backbone structure, rather than from a global energy surge.

This extension is conditional: it concerns degraded systems near the lower edge of the access region, not healthy systems already stably within it. Its empirical consequence is prediction P1, but direct biological validation of P1 is not claimed here.

Three-stage evaluation. The paper tests model-internal consistency in synthetic Kuramoto networks, applies the observables to public propofol EEG datasets plus sleep, ketamine, DoC, and dementia-proxy checks, and benchmarks GCC against spectral, artifact, lagged-coupling, phase-only, source-proxy, and reliability controls. These analyses establish operational state sensitivity, not direct selective-preservation validation in structurally degraded brains.

2. Core Intuition

The model assumes that conscious access is favoured neither by maximal desynchronization nor by maximal synchronization. Too little coordination prevents global integration; too much rigid coordination collapses the dynamics onto few inflexible modes. The favourable range lies

between these extremes, where global coordination, reduced but not collapsed dimensionality, and temporal variability co-occur [4, 8].

The core regime claim is therefore not “consciousness is synchrony” but:

Level of consciousness and global access arise in a regime of balanced consensus formation within a noisy, heterogeneous network.

3. Core Dynamical Model

We model N neural populations as phase oscillators on a weighted network. The dynamics of population i is given by

$$d\theta_i(t) = \left[\omega_i + K \cdot G(t) \cdot g_i(t) \sum_j W_{ij}(t) \sin(\theta_j(t) - \theta_i(t) - \varphi_{ij}(t)) \right] dt + \sigma_i(t) dB_i(t) \quad (1)$$

where:

- $\theta_i(t)$: phase of population i
- ω_i : intrinsic frequency
- K : global coupling scale
- $G(t) > 0$: global gain or excitability factor
- $g_i(t) \in [0, 1]$: population-specific preservation factor of afferent efficacy
- $W_{ij}(t) \geq 0$: normalized residual topology
- $\varphi_{ij}(t)$: effective phase delays
- $\sigma_i(t) \geq 0$: population-specific noise amplitude
- $B_i(t)$: independent standard Brownian motions

The matrix $W(t)$ is obtained from a raw residual adjacency $A(t)$ by row normalization:

$$W_{ij}(t) = \frac{A_{ij}(t)}{m_i(t)}, \quad m_i(t) = \sum_j A_{ij}(t), \quad (2)$$

78 provided $m_i(t) > 0$; otherwise $W_{ij}(t) = 0$.

79 The key structural move is the separation of topology W , global excitability G , local preser-
 80 vation g_i , and noise σ_i . Row normalization makes incoming structure comparable but would
 81 otherwise mask structural loss; g_i retains that loss explicitly. The phase-oscillator framework
 82 follows established work on synchronization thresholds in heterogeneous networks [14].

83 4. Degradation as Topology and Gain Loss

84 The GCC does not model degradation as pure edge removal. Structural and functional damage
 85 are separated as:

- 86 1. $A(t)$ or $W(t)$ describes which connections are topologically preserved.
- 87 2. $g_i(t)$ describes how strongly the remaining inputs remain functionally effective.

88 A simple working definition is

$$g_i(t) = \frac{m_i(t)}{m_i^{\text{ref}}}, \quad (3)$$

89 where m_i^{ref} denotes reference incoming mass and $m_i(t)$ current preserved mass. Thus row nor-
 90 malization of $W(t)$ does not erase afferent loss, which remains encoded in $g_i(t)$.

91 For degenerative conditions, the model takes the defensive form:

92 *In families of degraded networks, the operational access threshold typically shifts*
 93 *toward higher effective coupling, higher gain, or more favourable noise conditions.*

94 This is a model-internal, empirically testable tendency, not an unconditional universal claim.

Materials and Methods

5. Three Operational Observables

5.1 Global Coherence

$$R(t) = \frac{1}{N} \left| \sum_j e^{i\theta_j(t)} \right| \quad (4)$$

$R(t)$ measures macroscopic phase coherence. It is necessary for coordination but insufficient on its own, because rigid pathological coherence and flexible integration must be distinguished.

5.2 Effective Dimensionality

For a window of length τ_D , define the centred activity

$$X_i(s) = \cos(\theta_i(s)) - \langle \cos(\theta_i) \rangle_{[t-\tau_D, t]} \quad (5)$$

and the unregularized window covariance matrix $\hat{C}_D(t) = \text{Cov}(X_1, \dots, X_N)$. To avoid numerical problems in low-rank states, we use a scaled ridge regularization:

$$\varepsilon_D^\lambda(t) = \max\left(\lambda \frac{\text{tr}(\hat{C}_D(t))}{N}, \varepsilon_{\text{floor}}\right), \quad (6)$$

with a dimensionless parameter $\lambda > 0$ and an absolute floor $\varepsilon_{\text{floor}} > 0$. The regularized covariance matrix is

$$C_D^\lambda(t) = \hat{C}_D(t) + \varepsilon_D^\lambda(t) \cdot I. \quad (7)$$

The effective dimensionality is then the participation ratio:

$$D_{\text{eff}}^\lambda(t) = \frac{[\text{tr} C_D^\lambda(t)]^2}{\text{tr}[(C_D^\lambda(t))^2]} \quad (8)$$

Small values indicate compression onto few modes; large values indicate distributed activity.

The relevant range lies between fragmentation and rigid collapse.

5.3 Temporal Variability (Metastability)

For a window τ_M define:

$$M_\tau(t) = \text{Var}(R(s) : s \in [t - \tau_M, t]) \quad (9)$$

M_τ measures temporal variability of global coherence. It becomes interpretable only together with R and D_{eff} .

6. Operational Access Region

The GCC does not define an ontological criterion for consciousness but an operational access region in the space of observables. Relative to a pre-specified calibration, window choice, and regularization, a system is internally deemed *access-compatible* if

$$\boxed{R(t) > R_{\min}, \quad D_{\min} < D_{\text{eff}}^\lambda(t) < D_{\max}, \quad M_{\min} < M_\tau(t) < M_{\max}} \quad (10)$$

These inequalities are not a metaphysical definition of consciousness. They are a data- and scale-relative working description of a dynamical range favourable for global access.

6.1 Multiscale Robustness Formulation

Let $\mathcal{T}_D$, $\mathcal{T}_M$, and Λ be finite sets of admissible window lengths and ridge parameters, respectively.

For each triple $(\tau_D, \tau_M, \lambda)$ define the indicator

$$\mathcal{A}(t; \tau_D, \tau_M, \lambda) = \begin{cases} 1, & \text{if all three inequalities (10) are satisfied,} \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

The multiscale access index is

$$\Pi(t) = \frac{1}{|\mathcal{T}_D| |\mathcal{T}_M| |\Lambda|} \sum \mathcal{A}(t; \tau_D, \tau_M, \lambda). \quad (12)$$

123 A state is window-robustly access-compatible if $\Pi(t) \geq q^*$ for a pre-specified fraction q^* .

124 6.2 Data-Driven Calibration of Regime Boundaries

125 Regime boundaries are estimated from a reference ensemble $\mathcal{H}$ of access-compatible segments.

126 For a pre-chosen $\alpha \in (0, 0.5)$:

$$\begin{aligned} R_{\min} &= Q_{\alpha}(R \mid \mathcal{H}), \\ D_{\min} &= Q_{\alpha}(D_{\text{eff}}^{\lambda} \mid \mathcal{H}), & D_{\max} &= Q_{1-\alpha}(D_{\text{eff}}^{\lambda} \mid \mathcal{H}), \\ M_{\min} &= Q_{\alpha}(M_{\tau} \mid \mathcal{H}), & M_{\max} &= Q_{1-\alpha}(M_{\tau} \mid \mathcal{H}). \end{aligned} \tag{13}$$

127 6.3 Two Positive Reference Levels: $\mathcal{H}_{\text{full}}$ and $\mathcal{H}_{+}$

128 For degradation analyses, GCC distinguishes two positive reference levels:

- 129 • $\mathcal{H}_{\text{full}}$: healthy or near-healthy access segments with high macroscopic integration.
- 130 • $\mathcal{H}_{+}$: broader positive ensemble including partial access segments (mildly sedated, mildly
- 131 degraded).

132 With an exclusion ensemble $\mathcal{U}$, this separates two claims:

- 133 • **full-access-compatible:** $\Pi_{\text{full}}(t) \geq q_{\text{full}}$
- 134 • **partial-re-entry-compatible:** $\Pi_{+}(t) \geq q_{+}$, even if $\Pi_{\text{full}}(t) < q_{\text{full}}$

135 This prevents partial re-entry from being falsely rejected merely because it is not healthy-like
136 full access.

137 7. Operational Access Threshold Instead of a Universal K_c

138 For finite, noisy, and topologically heterogeneous networks, K_c is not a universal single number.

139 The GCC uses an operational access threshold K_c^{op} :

$$K_c^{\text{op}} = \arg \max_K \frac{d\langle R \rangle}{dK} \quad \text{or} \quad K_c^{\text{op}} = \arg \max_K \chi_R(K), \quad \chi_R(K) = \text{Var}_t(R(t; K)). \quad (14)$$

The threshold is therefore treated as an observable regime transition rather than an a priori constant [14].

8. Basic Mathematical Properties

Under $W_{ij} \geq 0$, $\sum_j W_{ij} = 1$, and $|\sin(\cdot)| \leq 1$, the deterministic coupling term is bounded:

$$|K \cdot G(t) \cdot g_i(t) \sum_j W_{ij}(t) \sin(\cdots)| \leq K \cdot G(t) \cdot g_i(t). \quad (15)$$

For the observables the elementary bounds hold:

$$1 \leq D_{\text{eff}}(t) \leq \text{rank}(C_D(t)) \leq N, \quad (16)$$

If $R(s)$ is constant over the window, then $M_\tau(t) = 0$. High coherence alone is therefore insufficient; a rigid state has low metastability.

9. Selective-Preservation Re-Entry as a Degradation Extension

The GCC contains a degradation extension: after exit from the access-compatible regime, selective preservation of efficacy on a residual subnetwork may permit transient partial re-entry. This is the most distinctive mechanistic extension, but the evaluated core of the paper is P2–P6 in simulations and public EEG/PSG validation.

Lucidity-like re-entry remains a possible future clinical application, not an empirical result established here.

9.1 Residual Backbone Structure

Let S be a representation-relevant population subset. “Residual” means not healthy, but connected enough to support a scaffold:

$$S(t; g_*) = \text{largest connected component of the subgraph on } \{i : g_i(t) \geq g_*\}. \quad (17)$$

9.2 Selective Gain Preservation Instead of a Global Energy Surge

The proposed extension mechanism is:

A lucidity-like or re-integration episode, if produced by the GCC mechanism, would not correspond to a global energy surge but to short-term functional concentration of remaining efficacy on a residual backbone structure.

The effective afferent drive of population i is proportional to

$$\Lambda_i(t) = K \cdot G(t) \cdot g_i(t). \quad (18)$$

Even if $G(t)$ decreases overall, preserved populations can gain relative influence over collapsing populations, yielding temporary residual-backbone reorganization.

9.3 Three Conditions for a Re-Entry

Selective-preservation re-entry requires:

1. **Structural residual condition:** A residual backbone structure S is still present.
2. **Gain condition:** For populations in S , $\Lambda_i(t)$ remains sufficiently high during a short interval.
3. **Dynamical admissibility:** Noise, delay dispersion, and frequency spread prevent neither integration nor flexible dynamics.

These conditions are strict; the model does not imply a final clarity episode in every degraded or dying process.

9.4 What the Model Explicitly Does Not Claim

The model does *not* claim that terminal or paradoxical lucidity is demonstrated here, that lucidity has a single mechanism, that noise reduction is sufficient, that high coherence explains

autobiographical memory, or that dying processes necessarily produce a final consensus-like peak.

10. Predictions

P2–P6 form the evaluated prediction family. P1 is retained as a clinical degradation extension, model-internally instantiated by P6 but not biologically validated here.

P2. Critical range. Near the access threshold the system shows non-linear reorganization rather than arbitrary gradual drift.

P3. Degradation shift. In degraded network ensembles, K_c^{op} shifts toward higher effective coupling or gain.

P4. Bounded interval. Global access is expected in an intermediate range, not at maximal desynchronization or rigid synchrony.

P5. State trajectories. State transitions can be described as motion in the space spanned by W , G , g_i , σ_i , and φ_{ij} .

P6. Model re-entry. Under matched degradation, selective preservation on a residual sub-network should yield higher Π than uniform decline and should not reduce to generic retuning.

P1. Clinical re-integration extension. In severely degraded biological systems, reduced overall activity but elevated relative residual-subnetwork integration should correlate with improved access. This requires future direct validation.

11. Empirical Falsifiability

The model fails if robust contrary findings show that:

- conscious access occurs without macroscopic integration;
- severe degradation does not shift the operational access region;
- rigid high coherence outperforms an intermediate range;

- selective preservation performs no better than uniform decline;
- or signatures disappear under artifact and robustness controls.

Thus GCC fails if calibrated regime boundaries show no reliable discriminatory power.

Results

The following sections report the model-internal evaluation of predictions P2–P6 on synthetic Kuramoto networks (Section 12), the application of the GCC observables to public EEG/PSG datasets across propofol sedation, sleep, and disorders of consciousness (Section 13), and spectral/artifact/source level stress tests (Section 13.4). P1 is discussed as a clinical extension and future empirical target, not as a biological result established here.

12. Numerical Validation on Synthetic Networks

To verify the internal consistency of the model and the formal correctness of the predictions in Section 10, we implement the full dynamical system on synthetic networks and test the evaluated prediction family P2–P6 under controlled conditions. P1 is not treated as biologically validated by these simulations. Rather, P6 provides a model-internal instantiation of the selective-preservation mechanism that P1 would require in structurally degraded clinical systems. This section establishes that (i) the predicted signatures follow from the model equations, (ii) the observables reliably distinguish regimes, and (iii) selective-preservation re-entry is reproducible in simulated data on an ensemble level. This is a *model-internal evaluation*, not an empirical validation against biological degradation data.

12.1 Calibration of Regime Boundaries

We realize the core dynamical model (Section 3) on Watts–Strogatz networks with $N = 100$ nodes, mean degree 10 and rewiring probability 0.1. The coupling matrix W_{ij} is binary. Integration proceeds via Euler–Maruyama with $dt = 0.01$ over $T = 100$ time units. Natural frequencies ω_i are drawn from a standard normal distribution; the gains g_i are initialized to 1 and modified only in specific experiments.

Calibration of the regime boundaries $R_{\min}$, $D_{\min}$, $D_{\max}$, $M_{\min}$, $M_{\max}$ follows the procedure in Section 6. We generate an ensemble of 12 networks under canonical parameters ($K = 2.5$,

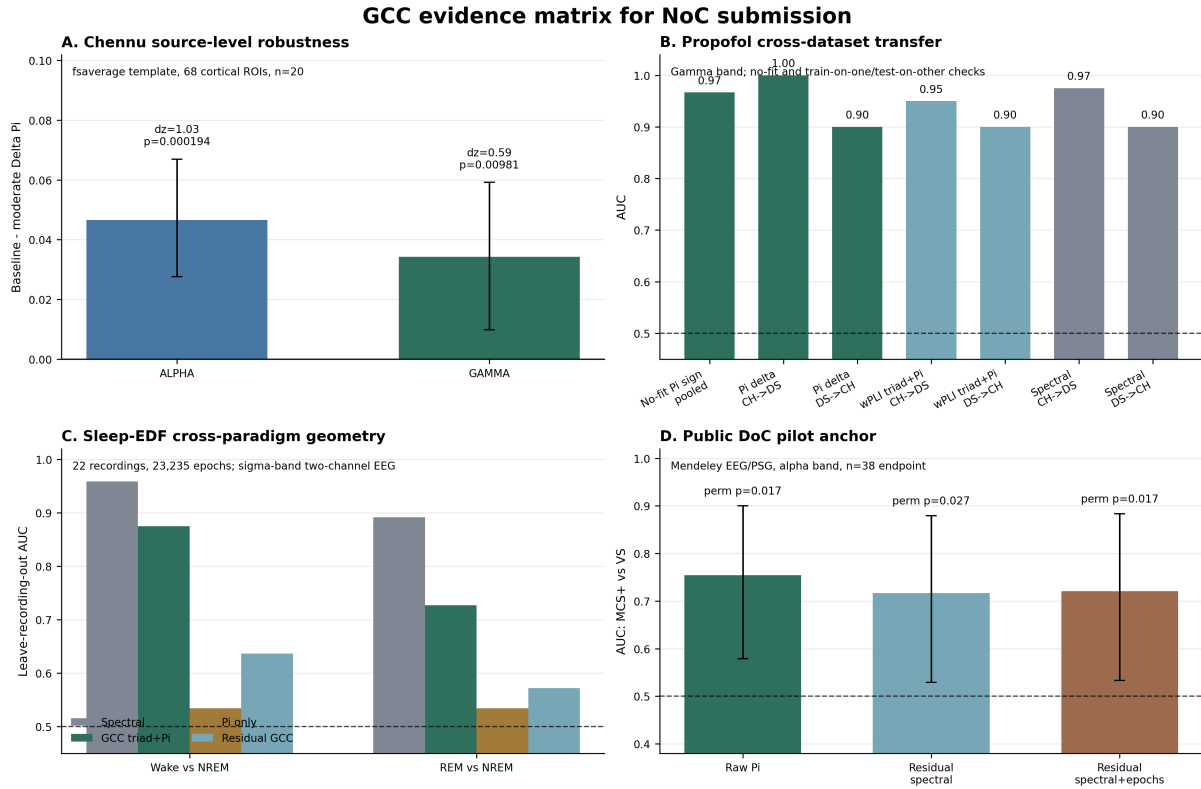


Figure 1: **NoC evidence matrix.** (A) Chennu template-source robustness: after mapping EEG to an fsaverage source model and reducing to 68 cortical ROIs, baseline II remains higher than moderate-sedation II in alpha and gamma bands. (B) Propofol cross-dataset transfer: no-fit and train-on-one/test-on-the-other gamma benchmarks remain above chance; spectral bandpowers are shown as a strong conventional baseline. (C) Sleep-EDF cross-paradigm state geometry: the GCC triad plus II separates Wake/NREM and REM/NREM above chance across 22 recordings, but spectral sleep features remain stronger and II alone is weak. (D) Public disorders-of-consciousness pilot anchor: alpha-band II distinguishes MCS+ from VS and remains positive after spectral and retained-epoch residualization.

Alt text: Four-panel evidence summary showing source-space propofol effects, cross-dataset propofol transfer, Sleep-EDF state classification, and a small disorders-of-consciousness pilot anchor. Across panels, GCC measures are reproducibly state-sensitive but are not stronger than all spectral baselines.

227 $\sigma = 0.2$) as the positive reference $\mathcal{H}_{\text{full}}$, extract the observables over a stationary window, and
 228 compute the $\alpha = 0.1$ quantiles. The lower bound M_{min} degenerates to zero in this ensemble con-
 229 figuration, indicating that some healthy systems lock into quasi-stable states. For all subsequent
 230 tests, $M_{\text{min}} = 0$ is used; the metastability condition thus effectively reduces to $M_{\tau} < M_{\text{max}}$.

231 12.2 P2 — Nonlinear Reorganization and Susceptibility Peak

232 Prediction P2 states that the access indicator II should increase nonlinearly with the coupling
 233 strength K along a sigmoidal curve rather than a linear one. We sweep $K \in [0, 4]$ and fit
 234 both a sigmoid and a linear function to the resulting $\Pi(K)$ curve. The sigmoidal fit achieves

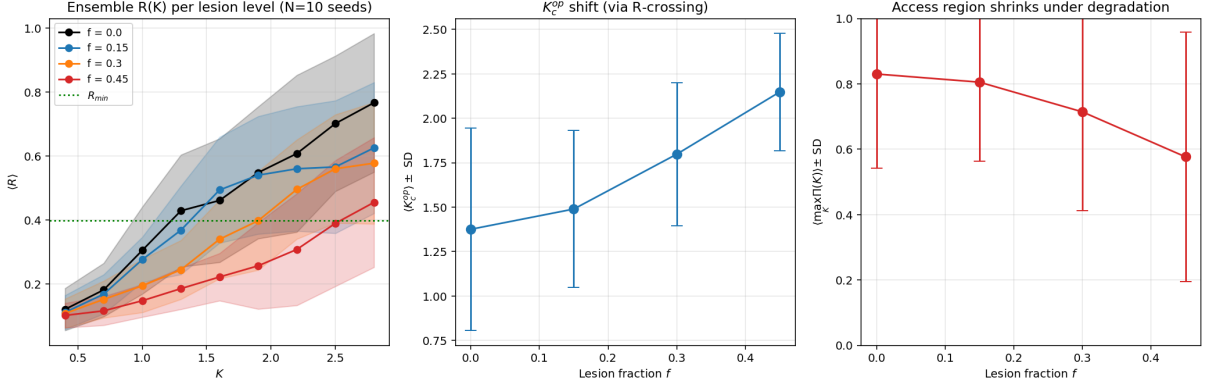


Figure 2: P3 and P4 Ensemble ($N_{\text{seeds}} = 10$): Left — $\langle R \rangle$ vs. K for four degradation levels, shaded bands $\pm$ SD; horizontal line marks $R_{\min}$. Middle — shift of $\langle K_c^{\text{op}} \rangle$ with lesion fraction, error bars SD. Right — maximum access index $\langle \max_K \Pi(K) \rangle$ vs. lesion fraction.

Alt text: Three plots summarize synthetic degradation sweeps: coherence curves shift with lesion fraction, the operational coupling threshold increases as degradation grows, and the maximum access index decreases with stronger lesioning.

$R^2 = 0.982$, the linear fit $R^2 = 0.948$; $\Delta R^2 = 0.034$. The curvature is clearly present but not abrupt, consistent with the “nonlinear but not strictly discontinuous” character that the model predicts. A susceptibility peak (standard deviation of R across network realizations) is visible in the midrange of the transition. P2 is confirmed.

12.3 P3 and P4 — Shift of Access Threshold and Bounded Interval (Ensemble)

Predictions P3 and P4 concern the behavior of the access threshold under degradation. We introduce graded degradation by removing a fraction $f \in \{0, 15, 30, 45\}\%$ of network edges (topological degradation). For each (K, f) combination, we run $N_{\text{seeds}} = 10$ independent network seeds and record K_c^{op} (first K -crossing of $R_{\min}$) and $\max_K \Pi(K)$ per seed, then average over the ensemble; detailed values are provided in Supplementary Table S1.

P3 is confirmed by the monotonic upward shift of $\langle K_c^{\text{op}} \rangle$: with increasing lesion the system requires stronger coupling to reach the access region. Pairwise Wilcoxon tests between lesion levels show a significant shift for $f = 0.00$ vs. $f = 0.45$ ($p = 0.002$). P4 is confirmed by the bounded maximum of $\Pi(K)$: even at optimal K , no state reaches $\Pi = 1$, and the attainable maximum fraction decreases monotonically from $\langle \max \Pi \rangle = 0.83$ (intact) to 0.58 (45% lesion), a total reduction of -30% (Figure 2). Empirically this means: the access region is a bounded range, not an open half-space, and it quantitatively shrinks under increasing degradation.

12.4 P5 — Anesthesia Trajectory in Parameter Space (Ensemble)

Prediction P5 describes the expected trajectory under a dynamic anesthesia protocol: the state should decrease smoothly during induction, collapse at maximal effect, and recover upon washout. To capture the ensemble-statistical nature of Π correctly, the trajectory was repeated with $N_{\text{seeds}} = 20$ independent network seeds; reported values are means and standard deviations across the seed ensemble. We implement a time-varying gain $G(t)$ with schedule: baseline ($G = 1.0$, $t < 3\text{s}$), induction ($G : 1.0 \rightarrow 0.2$), deep ($G = 0.2$), emergence ($G : 0.2 \rightarrow 1.0$), recovery ($G = 1.0$). The ensemble-averaged access indicator follows the expected course: baseline $\langle \Pi \rangle = 0.47 \pm 0.41$, deep $\langle \Pi \rangle = 0.00 \pm 0.00$, recovery $\langle \Pi \rangle = 0.46 \pm 0.29$. The trajectory in (R, D_{eff}) space leaves the access region and re-enters it smoothly, without jumps (Figure 3). P5 is confirmed: state transitions are motion in parameter space, not jumps between discrete states. The substantial standard deviation in the active phases (baseline, recovery) is a finding in its own right: Π on individual stochastic realizations of a Kuramoto network is strongly seed-dependent; its semantics as an *expectation-value-stable regime indicator* becomes clear only in the ensemble. The deep anesthesia phase shows the expected unambiguous signature ($\Pi = 0$ for all seeds).

12.5 P6 — Selective-Preservation Re-Entry on a Residual Backbone (Ensemble)

P6 tests the selective-preservation mechanism inside the model system. The system is placed in severe collective degradation, and we test whether a small subnetwork S (30 nodes out of 100), in which the gains are selectively boosted ($g_i : 0.5 \rightarrow 0.8$) while all non- S gains simultaneously collapse ($g_i : 0.5 \rightarrow 0.1$), can transiently reach the access region. The comparison condition is a uniform gain decay in which all g_i drop by the same factor. Both conditions follow four phases (baseline, global decline, re-entry window, final collapse). This protocol is not a biological validation of P1; it is a controlled instantiation of the mechanism that P1 would later require in structurally degraded clinical data.

To capture the ensemble-statistical nature of Π correctly, both conditions were simulated pairwise with $N_{\text{seeds}} = 20$ independent seeds. Reported values are ensemble means and standard deviations plus a paired statistical test. Results are summarized in Figure 4; detailed values are provided in Supplementary Table S2.

The quantitative difference in the re-entry window $\langle \Pi_S^B \rangle - \langle \Pi_S^A \rangle = 0.122$ is significant by paired

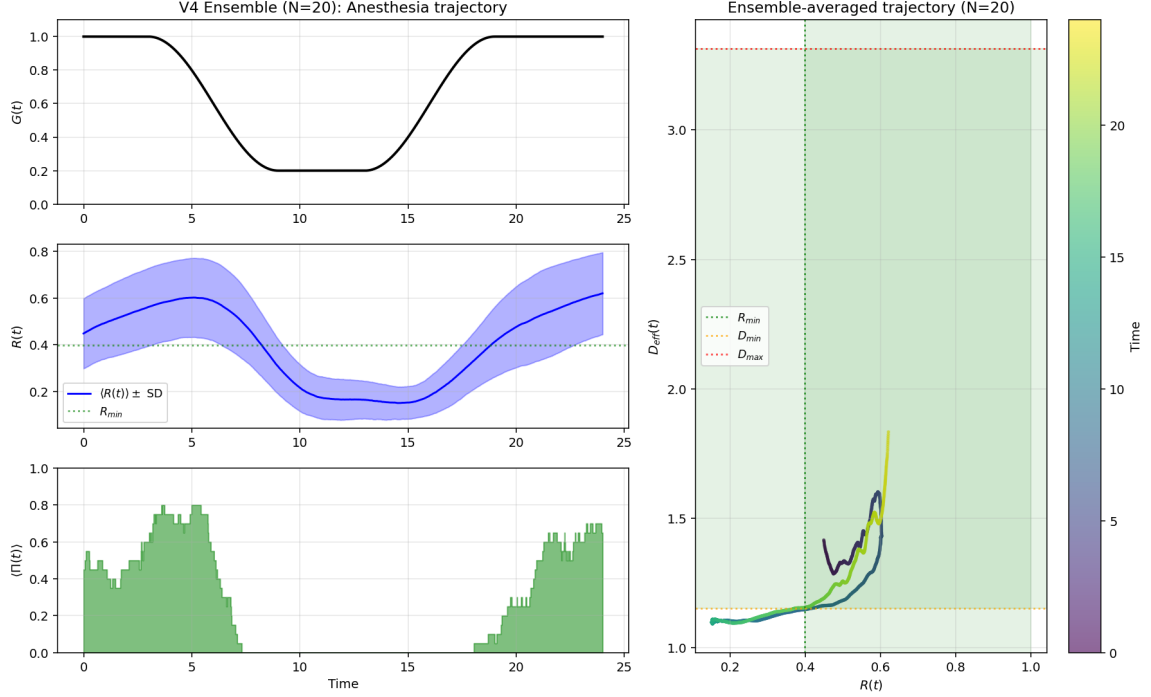


Figure 3: P5: Ensemble-averaged anesthesia trajectory ($N_{\text{seeds}} = 20$). Left top to bottom: $G(t)$ schedule, $\langle R(t) \rangle \pm \text{SD}$, ensemble-averaged $\langle \Pi(t) \rangle$. Right: ensemble-averaged trajectory in (R, D_{eff}) space, color-coded by time; shaded region marks the access region. The trajectory runs smoothly through the accessible region, leaves it in the deep phase, and returns through emergence.

Alt text: Synthetic anesthesia simulation in which gain changes over time, coherence and access index fall during the deep phase, and the state-space trajectory leaves and later re-enters the calibrated access region.

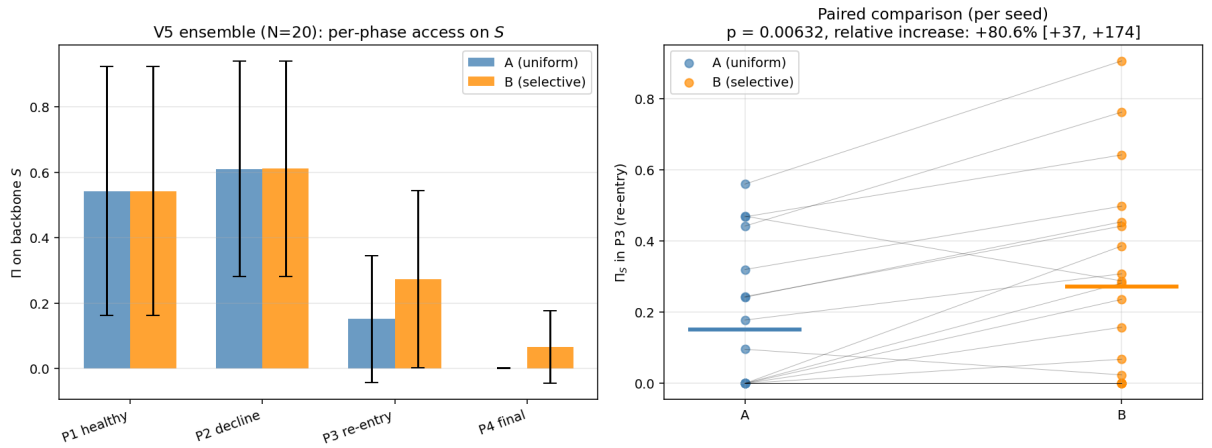


Figure 4: P6: Ensemble results of the selective-preservation re-entry simulation ($N_{\text{seeds}} = 20$). Left: phase-wise Π_S in condition A (uniform, blue) and condition B (selective preservation, orange); error bars show ensemble SD. Right: paired comparison across seeds in phase 3 (re-entry); each line connects a seed pair. Of 20 seeds, 16 show $\Pi_S^B \geq \Pi_S^A$ and 4 show $\Pi_S^B < \Pi_S^A$; paired Wilcoxon $p = 0.006$.

Alt text: Bar and paired-line plots comparing uniform degradation with selective preservation. The selective-preservation condition shows higher backbone access during the re-entry phase for most simulation seeds.

t-test ($t = 3.57$, $p = 0.002$) and paired Wilcoxon ($W = 9.0$, $p = 0.006$). The relative increase is +81% (bootstrap 95% CI: [+37%, +174%]), Cohen’s $d = 0.82$ (large effect). Sixteen of 20 seeds show $\Pi_S^B \geq \Pi_S^A$; in phase 4, condition B retains a marginal residual signal on S ($\langle \Pi_S^B \rangle = 0.07$ vs. $\langle \Pi_S^A \rangle = 0.00$), consistent with a short-lived gain residuum. Crucially, this effect occurs *while overall gain decreases* — it arises not from a global energy surge but from selective preservation of efficacy on a subset.

12.6 P6 Retuning Control: Separating Re-Entry from Retuning

A second concern deserves explicit testing. In the initial P6 simulation, $\langle \Pi_{\text{full}}^A \rangle$ rises from 0.53 (baseline phase) to 0.69 (decline phase) in *both* conditions. Uniform gain decline alone therefore improves the access index, creating a possible retuning confound for the selective-preservation mechanism.

To separate retuning from re-entry, the P6 protocol was repeated at four baseline coupling strengths $K_{\text{baseline}} \in \{1.6, 2.0, 2.5, 3.0\}$ with $N_{\text{seeds}} = 20$ each (Figure 5; Supplementary Table S3). Retuning and selective re-entry were quantified separately per K_{baseline} .

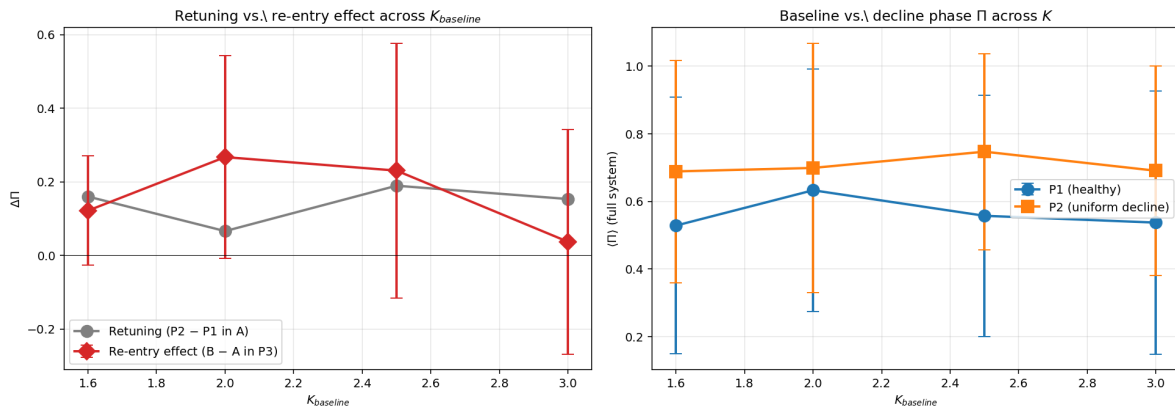


Figure 5: P6 retuning control. Left: retuning effect (gray) and re-entry effect (red) across baseline coupling strength. At $K = 2.0$, both effects are cleanly separated (retuning +0.07, re-entry +0.27, factor 4); at $K = 3.0$ the re-entry effect vanishes (ceiling). Right: full-system Π in baseline and uniform-decline phases across K ; shows that retuning occurs broadly across the K -range but stays in the single-digit to mid-range.

Alt text: Two control plots show that generic retuning effects are smaller than selective re-entry effects at the main coupling value, while ceiling effects remove the re-entry contrast at high baseline coupling.

Three descriptive findings follow from this sweep. *First:* the re-entry effect (B over A on S) is significant at three of the four K values. *Second:* retuning and re-entry are separate quantities at all K , with the cleanest dissociation at $K = 2.0$ (retuning +0.07, re-entry +0.27, factor 4). *Third:* at $K = 3.0$ the re-entry effect collapses to ceiling level (+5%, n.s.), while retuning persists

— consistent with the model, since selective preservation only acts where the full system exits the access region.

Primary analysis: pooled sweep statistic. To avoid post-hoc selection, we report the entire K -sweep as the main result through a pooled regression with seed clustering. The model $\Pi \sim \text{Condition} \cdot K_{\text{baseline}} + (1|\text{Seed})$ over all 80 paired observations (20 seeds $\times$ 4 K values) yields a highly significant main effect of Condition (B vs. A), $\hat{\beta}_{\text{Cond}} = +0.164$, cluster-robust SE = 0.040, $p = 3.6 \times 10^{-5}$. The $\text{Condition} \times K$ interaction is not significant ($\hat{\beta}_{\text{Int}} = -0.069$, $p = 0.32$), i.e., there is no statistical evidence that the size of the re-entry effect depends systematically on K — with the exception of the descriptively visible ceiling at $K = 3.0$, which does not reach interaction significance at the available sample size (4 K values).

Secondary, descriptive plausibility check. As a complementary, non-primary statistic, we report a pooled paired Wilcoxon test across all 80 pairs ($W = 353.5$, $p = 5.5 \times 10^{-7}$, Cohen’s $d_{\text{paired}} = 0.56$). We explicitly note that this test pools non-independent observations (each seed appears four times, once per K value) and therefore serves only as a descriptive consistency check on the main effect, not as standalone inference. The primary analysis, which correctly accounts for the dependency structure, is the mixed regression with seed clustering.

Interpretation. The re-entry effect is *not confined to a favorably chosen K_{baseline}* : it is robust across the sweep ($p < 10^{-6}$ pooled), significant in three of four individual K values, and its mean magnitude is not adequately captured by a simple retuning explanation. The result at $K = 2.0$ (+102%, $p = 0.0009$) is retained in the paper, but as a *point of maximum sensitivity* within the sweep, not as a primary result selected from the sweep. The a priori mechanistic rationale: the calibration ensemble $\mathcal{H}_{\text{full}}$ was generated at $K = 2.5$; $K = 2.0$ (slightly below) is the point at which the full system barely exits the access region under decline, while backbone S is not yet at ceiling — hence the range of greatest diagnostic sensitivity. The re-entry effect there is consistent with the robust mechanism documented across the entire sweep.

The core claim of P6 — selective gain preservation enables a transient re-entry on the residual backbone that exceeds mere retuning — is thus supported and hedged against the post-hoc selection objection: the effect is documented as a *sweep-level phenomenon*, not as a single finding at a favorably chosen parameter.

12.7 Assessment of the Numerical Validation

The evaluated prediction family P2–P6 is directly testable on synthetic networks. P1 receives only model-internal support: P6 shows that the required selective-preservation mechanism can occur under controlled assumptions, but the biological EEG analyses do not test selective gain preservation in severely degraded systems.

- **P2:** Nonlinear reorganization ($\Delta R^2 = 0.034$ between linear and sigmoidal fit), not strictly discontinuous.
- **P3:** K_c^{op} (via dR/dK) shifts monotonically from 1.40 (intact) to 2.80 (45% lesion).
- **P4:** $\langle \max_K \Pi(K) \rangle$ stays below 1 and decreases monotonically with increasing degradation from 0.83 (intact) to 0.58 (45% lesion), total reduction -30% ; both quantities (threshold shift and bounded maximum) behave robustly at the ensemble level.
- **P5:** Ensemble-averaged trajectory in (R, D_{eff}) space runs smoothly; $\langle \Pi \rangle$ follows the anesthesia–recovery cycle (baseline 0.47, deep 0.00, recovery 0.46).
- **P6:** Differential degradation produces ensemble-level transient re-entry on the backbone. Across the retuning-control sweep $K \in \{1.6, 2.0, 2.5, 3.0\}$, Condition has a significant effect (cluster-robust $\hat{\beta}_{\text{Cond}} = +0.164$, $p = 3.6 \times 10^{-5}$, $d_{\text{paired}} = 0.56$), maximal at $K = 2.0$ ($+102\%$, $p = 0.0009$). This supports the model-internal P1 mechanism but does not validate P1 biologically.

Ensemble-statistical nature of Π . In the simulations, Π is an expectation-value-stable ensemble quantity: individual stochastic realizations scatter strongly around the ensemble mean. The biological analyses therefore use a different operational strategy, calibrating Π within each participant’s own baseline recording. This bridge rests on an ergodicity-type working assumption, discussed explicitly in Section 13.6.

Limits of this validation. The synthetic networks are Watts–Strogatz rather than biological connectomes; calibration and testing occur within the same ensemble paradigm; regime boundaries depend on quantile and window choices; and P6 uses a fixed backbone size and boost protocol. In addition, the lower metastability bound degenerates under the present calibration ensemble, making the implemented access score closer to a two-dimensional (R, D_{eff}) criterion

with an upper variability constraint. These limits do not invalidate the internal test, but they define what must be tightened in future connectome-based and out-of-sample simulations.

The simulations thus evaluate the model as *internally consistent and functional*: the dynamics produce the predicted signatures, the observables separate regimes, and selective-preservation re-entry is reproducible at the ensemble level. The next step — application to real EEG data — is carried out in the following section as a public-dataset validation.

13. Public EEG Validation Across State and Clinical Domains

The biological analyses tested whether GCC observables respond reproducibly to public EEG/PSG state changes. They include two independent propofol datasets, Sleep-EDF, a public DoC EEG/PSG pilot, ketamine, and a dementia-proxy dataset. They do not directly validate P1, because no dataset contains documented selective-preservation or lucidity-like re-entry.

13.1 Chennu Propofol Pipeline

The primary pipeline used the public Chennu et al. propofol EEG dataset [2]: 20 healthy participants, 91 channels, 250 Hz, and baseline, mild, moderate, and recovery resting states. Data were filtered, average-referenced, Hilbert-transformed in gamma (30–45 Hz) and alpha (8–15 Hz), calibrated within each participant’s baseline recording ($\alpha = 0.1$, $\tau_D = 200$ ms, $\tau_M = 500$ ms), and scored under those individual bounds. Baseline-vs-moderate effects were $\Delta\Pi = 0.111$ in gamma (95% CI [0.086, 0.137], Wilcoxon $p = 1.9 \times 10^{-4}$, $d = 0.99$) and $\Delta\Pi = 0.065$ in alpha (95% CI [0.038, 0.093], $p = 0.0023$, $d = 0.69$). Sensitivity testing over 36 calibration/window configurations per band showed 100% directional stability.

13.2 Template-Source Robustness

To test sensor-level inflation, Chennu was recomputed after template-source projection to 68 cortical ROI time series. The decline survived: alpha $\Delta\Pi = 0.0466$ (95% CI [0.0276, 0.0670], $d_z = 1.03$, $p = 1.94 \times 10^{-4}$) and gamma $\Delta\Pi = 0.0342$ (95% CI [0.0098, 0.0592], $d_z = 0.59$, $p = 0.0098$; Figure 1A). This does not replace individualized source reconstruction, but it rules out a purely raw-sensor finding.

13.3 Independent Propofol Replication

OpenNeuro DS005620 provided an independent propofol replication with 21 healthy participants and repeated resting sedation recordings [1]. Using eyes-closed awake recordings for calibration ($N = 20$ paired participants), gamma Π declined by 0.387 for awake-vs-sed (95% CI [0.305, 0.479], $d = 1.91$, FDR sign-flip $q = 1.0 \times 10^{-4}$) and 0.341 for awake-vs-sed2 (95% CI [0.267, 0.416], $d = 1.97$, $q = 1.0 \times 10^{-4}$). Alpha effects were smaller but robust. Pooled across Chennu and DS005620, participant-cluster bootstrap estimated $\Delta\Pi = 0.187$ (95% CI [0.148, 0.228]; Wilcoxon over participant means $p = 5.0 \times 10^{-11}$). DS005620 effects are treated as dataset-specific separability, not universal effect size.

13.4 Spectral, Artifact, and Lagged-Coupling Controls

GCC was benchmarked against theta, alpha, beta, gamma, alpha/gamma ratio, and spectral entropy. Raw gamma Π was strong within Chennu (AUC = 0.87) and DS005620 (AUC = 1.00), but bandpower residualization reduced performance (AUC = 0.61 and 0.68); GCC is therefore not claimed as a generally bandpower-independent biomarker. DS005620 gamma effects survived posterior/centro-posterior channel restrictions while posterior 70–110 Hz power did not distinguish awake from sedation. A wPLI-style lagged-coupling proxy retained sedation sensitivity in both datasets [16]. Frozen cross-dataset transfer also generalized (standard gamma Π : pooled no-fit AUC = 0.967; Chennu→DS005620 AUC = 1.00; DS005620→Chennu AUC = 0.90), while spectral baselines remained competitive.

13.5 Sleep, DoC, Ketamine, and Degradation-Proxy Checks

Sleep-EDF provided a non-propofol state check [10]. In 22 PSG–hypnogram recordings (23,235 epochs), the sigma-band GCC triad plus Π separated Wake from NREM with AUC = 0.875 and REM from NREM with AUC = 0.727 (Figure 1C). Π alone was weak (AUC $\approx$ 0.534), supporting a multivariate regime-geometry interpretation rather than a scalar sleep biomarker. Because only two EEG channels are available, this is not a direct access validation.

The public DoC EEG/PSG dataset contained 42 EDF files: 30 VS/UWS, 4 MCS-, and 8 MCS+ recordings. The pre-specified alpha-band endpoint, MCS+ versus VS, was positive unadjusted but marginal across the exploratory family: raw Π AUC = 0.754 (95% CI [0.579, 0.900], permutation $p = 0.017$), spectral-residual AUC = 0.717 ($p = 0.027$), and spectral/epoch-count residual

AUC = 0.721 ($p = 0.017$; Figure 1D). The family-level FDR q-value was approximately 0.084. This is a pilot clinical anchor, not diagnostic validation.

Additional checks reduced implementation-specific freedom. Amplitude-normalized phase-only GCC retained propofol sensitivity (Chennu gamma $\Delta\Pi = 0.463$, $d = 2.37$; DS005620 gamma $\Delta\Pi = 0.439$, $d = 1.99$), as did a CSD/source-proxy pipeline. Ketamine shifted gamma Π downward while alpha D_{eff} increased, consistent with altered-state rather than propofol-like geometry [9]. DS004504 GCC features separated dementia-proxy participants from controls (AUC = 0.769) and correlated with MMSE [12]. A frozen validation specification records bands, windows, calibration, baselines, controls, and forbidden claims.

13.6 Bridge and Limits

In simulations, Π is an ensemble quantity; in EEG, it is calibrated within a participant’s baseline recording. This rests on an ergodicity-type working assumption supported pragmatically by two propofol datasets, split-half checks, and repeated-run analyses, but not derived from the model. The public validation remains limited: baseline Π is calibration-dependent, most analyses are channel-level, Hilbert channel phases approximate but do not equal model phases, DS005620 effect sizes exceed Chennu, residualized transfer is asymmetric, and GCC overlaps substantially with spectral markers. Most importantly, the data do not directly test P1. The present work establishes operational state sensitivity and model-internal mechanism, not clinical selective-preservation re-entry.

Discussion

The following sections discuss the relative originality of the GCC framework, the main discussion and interpretation of the findings, and a concluding summary with explicit boundaries of what the present work delivers and what it does not.

14. Relative Originality

The GCC claims no absolute monopoly on its components. Integration, synchrony, critical dynamics, global availability, and metastability are each well established in the literature [4, 5, 8, 15]. The relative distinctive feature lies in their explicit assembly into a single operational regime model for access-compatible neural dynamics, evaluated through a prediction family over

nonlinear transitions, degradation shifts, bounded access intervals, anesthesia-like trajectories, and selective-preservation re-entry in controlled simulations. The clinical degradation hypothesis is an extension of this regime framework, not the sole foundation of the manuscript.

15. Discussion

The main claim evaluated in this paper is that access-compatible dynamics occupy a bounded, calibrated regime rather than increasing monotonically with any single observable. The empirical part now supports this claim at the level appropriate for a principled computational model: GCC regime occupancy can be computed from real EEG, decreases reproducibly under propofol sedation across two independent public datasets, survives a gamma-band topographic artifact stress test, persists under spectrally closest run selection, transfers under frozen train-on-one-dataset/test-on-the-other rules, and remains present when zero-lag-sensitive coherence is replaced by a wPLI-style lagged-coupling proxy. At the same time, the spectral residualization, phase-surrogate, and spectral-baseline results impose an important boundary: GCC is not established here as a bandpower-independent biomarker, and conventional bandpowers remain strong competitors. It should be interpreted as a multi-observable regime composite whose condition information partly overlaps with conventional spectral state markers. Embedded in the broader framework of balanced consensus formation, the strongest empirically defended reading of the model is:

Level of consciousness and global access are favored in a critical dynamical regime in which residual network topology, effective gain structure, noise, and delays jointly produce sufficient but non-rigid macroscopic integration.

The counterintuitive degradation extension is narrower: functional re-integration in damaged systems may arise from selective preservation and gain redistribution, not only from increased global activity. The present work supports this statement as a controlled model-internal mechanism. It does not establish it in structurally degraded biological cohorts, and it should not be presented as an empirical result about terminal lucidity or disorders of consciousness.

15.1 Future Directions

The most important next step is not a stronger philosophical claim but a stricter empirical test. Three developments would directly strengthen or falsify the GCC. First, regime boundaries

should be calibrated out of sample across heterogeneous EEG/MEG cohorts. Second, template-source robustness should be replaced by individualized source-level EEG/MEG or intracranial recordings where possible. Third, selective preservation should be tested prospectively in structurally degraded clinical cohorts with repeated behavioral access measures and longitudinal recordings.

16. Conclusion

The GCC is not a full model of consciousness but a consistent, operationalized, and testable principled model of access-compatible neural dynamics. Its present status is strongest as a regime model evaluated by simulations and public EEG validation: the observables are computable, state-sensitive, reliable at recording level, and capable of expressing bounded access rather than monotonic coherence. Across two independent public propofol datasets, subject-calibrated GCC regime occupancy declines reproducibly under sedation, and the Chennu effect survives a template-source ROI robustness check. Sleep-EDF, a public DoC EEG/PSG dataset, ketamine, and dementia-proxy analyses add boundary-domain checks in which the GCC geometry differentiates sleep states, carries a pilot clinical MCS+ vs VS signal, shifts under altered conscious pharmacology, and captures neurodegenerative group structure without being overinterpreted as direct access validation. The empirical result is strengthened by subject-level confidence intervals, FDR/permutation transparency, frozen cross-dataset transfer, spectrally closest run controls, gamma artifact controls, a wPLI-based lagged-coupling stress test, amplitude-normalized phase-only GCC, CSD/source-proxy GCC, template-source robustness, reliability analyses, and a frozen validation specification; it is limited by substantial overlap with conventional spectral bandpower, mostly channel-level measurement, small clinical samples, and the absence of a direct P1 clinical re-entry dataset. Its principal theoretical move remains the separation of topology and functional efficacy through the gain model. The selective-preservation account of re-entry in structurally degraded systems remains a clinical extension and future empirical target, not the central validated result of the present paper.

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Author Contributions

Mehmet Oktay Tugan: Conceptualization, methodology, software, formal analysis, investigation, validation, visualization, writing—original draft, writing—review and editing.

Ethics Statement

This study is a secondary analysis of publicly available, de-identified human EEG datasets and synthetic simulations. No new human participants were recruited and no new human or animal data were collected. Ethical approvals and informed consent were obtained by the original data-generating studies where applicable. No additional institutional ethics approval was required for the present secondary analysis.

Data Availability

The re-analysed datasets are publicly available: Chennu et al. EEG [2] (DOI: 10.17863/CAM.68959), OpenNeuro DS005620 [1], Sleep-EDF Sleep Cassette [10], the Mendeley disorders-of-consciousness EEG/PSG dataset (DOI: 10.17632/6wx4n25h4v.1), the Farnes et al. ketamine EEG data [9], and OpenNeuro DS004504 [12]. Raw third-party datasets are not redistributed. Author-generated simulation code, analysis scripts, derived result tables, figures, and reproducibility instructions are provided in the public GCC reproducibility repository at <https://github.com>.

`com/oktaytugan82/gcc_simulation`. The NoC preflight-clean submission package is archived as release `v2.0.1-noc-preflight` on Zenodo (version DOI: 10.5281/zenodo.19818741; concept DOI for the latest repository version: 10.5281/zenodo.19744969).

Competing Interests

The author has declared that no competing interests exist.

Supplementary Material

Supplementary Text S1 contains the mathematical well-posedness results, observable bounds, finite-horizon stability estimates, the sufficient selective-preservation re-entry condition, and the discussion of Gronwall conservativity. Supplementary code and derived analysis outputs are provided in the reproducibility package described in the Data Availability statement.

Supporting Information

S1 Code. GCC simulation and analysis code (`gcc_simulation_v2.0.1-noc-preflight.zip`). Complete Python codebase for the GCC dynamical model, per-subject EEG analysis pipeline, ensemble simulations, retuning-control sweep, sweep-level statistical analysis, spectral baseline comparison, bandpower residualization, DS005620 replication, gamma artifact-control scripts, phase-surrogate controls, wPLI lagged-coupling controls, frozen cross-dataset benchmarks, amplitude-normalized phase-only GCC, CSD/source-proxy GCC, template-source Chennu robustness analysis, Sleep-EDF state-differentiation analysis, public DoC pilot analysis, ketamine boundary-state analysis, DS004504 degradation-proxy analysis, split-half/repeated-run reliability analysis, frozen validation specification, and figure-generation scripts. Includes reproducibility instructions, derived result tables, manuscript source files, and the Batch 3–18 validation outputs reported here. The public repository is https://github.com/oktaytugan82/gcc_simulation; the NoC preflight-clean submission snapshot is release `v2.0.1-noc-preflight`, archived on Zenodo with version DOI 10.5281/zenodo.19818741.

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